

Layer 3: LLM learner

offline, run manually;
reads champion code,
scores, traces, lessons;
proposes candidates

Gate

AST security check,
restricted built-ins,
timeout; scored on fault
battery; promoted if better

Layer 2: supervisor

deterministic Python;
every decision interval:
anomaly flags and
adjusted setpoints

Plant

tank levels and pumps;
injected leak faults

Layer 1: PI/PID

fixed, never generated;
tracks active setpoints,
drives the pumps

*logged
telemetry
(offline)*

telemetry

setpoints

